Abin Anil

Backend Engineer | Python, FastAPI, Django & Async Systems
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SUMMARY

Backend Engineer with a strong Python foundation, building async REST and WebSocket services, PostgreSQL-backed APIs, and Dockerized deployments on GCP. Backed by 2+ years supporting a production distributed platform at 99.8% uptime.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, JavaScript, FastAPI, Django REST, REST APIs, WebSockets, AsyncIO, Pandas, NumPy

Databases: PostgreSQL, pgvector, SQL (debugging and fixes)

Cloud & DevOps: Docker, GCP Cloud Run, Jenkins CI/CD, Git, GitHub

Debugging & Observability: Postman, Swagger, cURL, Kibana, Grafana

AI Integration: LLM APIs (Gemini, Groq), LangGraph, RAG, MCP, Gemini Live API, Scikit-Learn, XGBoost

PROJECTS

AI-Powered Technical Interview Assistant Oct 2026
GitHub: github.com/abinanil000datascience/ai-interview-assistant

- Built a low-latency real-time service (FastAPI, WebSockets) streaming bidirectional audio to the Gemini Live API, with a 6-step state-machine prompt handling interruptions.
- Engineered an async pre-fetch pipeline (`httpx`, GitHub REST API) that loads a user's repos at session start: **zero mid-call network calls**.
- Fixed CPU bottlenecks and hallucination loops in local STT (`faster-whisper`) with Silero VAD, byte-alignment checks and `asyncio.Lock`: **latency ∼60 s to <2 s**.

Nexus Intelligence – Multi-Agent AI Trading System Sep 2026
GitHub: github.com/abinanil000datascience/nexus-ai-trading-engine

- Built a four-agent trading backend (LangGraph, MCP, async FastAPI) fusing RSI/MACD signals (Pandas) with Gemini news sentiment into real-time buy/sell verdicts, with a React dashboard.
- Moved agent execution to background threads to remove event-loop blocking: **3–5x throughput** and **p95 latency ∼2.4 s to <800 ms** (1,000 concurrent tick simulation).

Healthcare IT Ticket Resolution Engine (ITSM) Aug 2026
GitHub: github.com/abinanil000datascience/Healthcare_ITSM_Agent

- Built a RAG API with hybrid search (BM25 + HNSW in pgvector) on Dockerized FastAPI: **2.6x faster retrieval** than an exact L2 scan (10K tickets) with **>97% context precision/recall** (RAGAS).
- Served a fine-tuned Llama-3 (8B, QLoRA) locally, saving **$50/month** versus GPT-4o (2,500 tickets/month).

AI Career Platform – Smart Resume Matching Aug 2026
GitHub: github.com/abinanil000datascience/ai-career-platform

- Built a semantic resume-to-job matching API (LangGraph, embeddings, PostgreSQL/pgvector) on async FastAPI with Gemini and Groq LLMs, deployed on GCP Cloud Run.
- Added async exponential backoff (Tenacity), lifting concurrent 100-resume batches from 85% to **100% eventual success**.

Supply Chain Risk Predictor May 2026
GitHub: github.com/abinanil000datascience/supply-chain-risk-predictor

- Built a Django REST API serving delayed-freight predictions at **<40 ms p95**; raised F1 from **0.65 to 0.77** on a 97% on-time dataset (SMOTE, XGBoost).

PROFESSIONAL EXPERIENCE

Technical Support Engineer, Feathersoft Aug 2024 – Present
Distributed DNA diagnostics platform

- Maintained 99.8% uptime and data integrity for a distributed diagnostics platform with automated report workflows.
- Cut production errors by 22% by debugging API and data-ingestion failures (cURL, Postman, Swagger).
- Cut MTTR by 20% with SQL fixes; sped up resolution by 15% via Kibana and Grafana log analysis.
- Supported AI/ML projects, using Python, Pandas, and LLM APIs to analyze model, data, and API issues.
- Led client triage (99% CSAT, 120+ escalations).

EDUCATION

B.Tech, Computer Science & Engineering 2020 – 2024
University College of Engineering, Thodupuzha